Junghoon Chung

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Summary

PhD candidate in Industrial & Systems Engineering with a strong background in **UX research, human factors, and data-driven design**. Experienced in designing and executing end-to-end **mixed-methods research** including interviews, surveys, controlled experiments, and longitudinal field studies. Skilled at integrating qualitative insights with large-scale behavioral and sensor data to inform product and system design. Proven ability to translate complex findings into actionable recommendations for cross-functional teams. Seeking a UX Research internship to support evidence-driven product development at scale.

Education

University of Washington, Seattle — **Sep 2025 – Jun 2029 (Expected)**
Ph.D in Industrial and Systems Engineering, Dean's Fellowship Recipient

Virginia Tech, Blacksburg — **Aug 2023 – May 2025**
M.S in Industrial and Systems Engineering (Thesis Track), GPA 3.89/4.0

Yonsei University, Seoul — **Mar 2017 – Feb 2023**
B.S in Industrial Engineering, GPA 3.52/4.0, Student Council President

Experience

Graduate Research Assistant, ***Human Automation Systems Lab, University of Washington*** — **Sep 2025 – Present**

- Designed controlled experiments and user studies to evaluate attention, workload, and performance in complex task environments; collected multimodal behavioral and physiological data to understand user states during sustained interaction
- Generated insights for designing systems that support sustained attention and reduce cognitive overload; communicated findings to lab stakeholders and external collaborators

Graduate Research Assistant, ***Occupational Ergonomics and Biomechanics Lab, Virginia Tech*** — **Aug 2023 – May 2025**

- Led end-to-end mixed-methods research on how knowledge workers interact with computer systems and workplace technologies; conducted interviews, surveys, and behavioral observations alongside sensor-based data collection
- Synthesized qualitative feedback and quantitative data to identify usability barriers and contextual drivers of behavior; analyzed longitudinal data (3 weeks/participant) to understand patterns in productivity, comfort, and task engagement
- Translated findings into design recommendations for adaptive workplace systems; presented results to cross-functional research teams

Key Projects

Driver Experience & Workload in V2X Pedestrian Alert Systems | City of Bellevue / University of Washington
Designed and conducted a field study investigating how drivers perceive and respond to connected vehicle pedestrian alerts at crosswalks. Collected **physiological measures (E4 Empatica), NASA-TLX workload ratings, and vehicle kinematics**; synthesized objective and subjective data to surface usability insights and inform alert design in automated vehicle safety contexts.

Human Vigilance & Cognitive Load Modeling | University of Washington
Investigated how cognitive load and alertness influence performance during sustained tasks by integrating behavioral measures with **30+ physiological and contextual features across 10,000+ observations** per participant. Developed predictive models improving performance by 20%; generated insights for designing systems that better support attention and reduce cognitive burden in safety-critical environments.

Context-Aware Sit-Stand Intervention for Knowledge Workers | Virginia Tech
Conducted a longitudinal field study (3 weeks/participant; 100,000+ records) using **interviews, behavioral logs, and sensor data** to understand posture-switching behavior and contextual barriers during computer work. Applied user clustering to identify distinct behavioral profiles; findings informed the design of **adaptive, personalized workplace interventions** balancing health and productivity. Published at HFES 2025; two journal manuscripts in preparation.

Skills

Research Methods: Interviews, Surveys, Usability Testing, Longitudinal Field Studies, Experimental Design, ESM
Analysis: Statistical analysis, Thematic coding, Evidence synthesis, Python, SQL
Collaboration: Cross-functional teamwork, Technical communication, Stakeholder presentation

Awards

Dean's Fellowship, College of Engineering — **University of Washington, 2025–2029**
Social-Innovation Capability Video Contest 2nd Place – Data mining — **Yonsei University, 2022**
Merit-based Scholarship (Academic Excellence) — **Yonsei University, 2021**